

DATABASE MANAGEMENT SYSTEM LAB

ASSIGNMENT NO. 05 – [Name : Ridafatema Patel, Rollno. 63]

Title : Unnamed PL/SQL code block : Use of control structure and exception handling is mandatory.

Consider the following schema/tables :

1. Borrower(Rollno, Name, IssueDate, Book, Status)
2. Fine(Rollno, FineDate, Amount)

- A. Accept Roll_no and Name of Book from the user.
- B. Check the number of days (from date of issue).
- C. If days are between 15 to 30 then the fine amount will be Rs 5per day.
- D. If no. of days>30, per day fine will be Rs 50 per day and for days less than 30, Rs. 5 per day.
- E. After submitting the book, status will change from I to R.
- F. if condition of fine is true, then details will be stored into fine table
- G. Also handles the exception by named exception handler or user defined exception handler.



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ORACLE Database Express Edition
User: SYSTEM
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Autocommit Display 10 Save Run

CREATE TABLE BORROWERSS(
ROLLNO NUMBER,
NAME VARCHAR(30),
ISSUEDATE DATE,
BOOK VARCHAR(30),
STATUS VARCHAR(10)
);

CREATE TABLE FINESS(
ROLLNO NUMBER,
DATES DATE,
AMOUNT NUMBER
);

DECLARE
V_ROLLNO    NUMBER    := :ROLLNO;
V_NAME      VARCHAR2(30) := :NAME;
V_ISSUEDATE DATE      := TO_DATE(:ISSUEDATE, 'DD-MM-YYYY');
V_BOOK      VARCHAR2(30) := :BOOK;
V_STATUS    VARCHAR2(10) := :STATUS;
BEGIN
INSERT INTO BORROWERSS (ROLLNO, NAME, ISSUEDATE, BOOK, STATUS)
VALUES (V_ROLLNO, V_NAME, V_ISSUEDATE, V_BOOK, V_STATUS);

DBMS_OUTPUT.PUT_LINE('RECORD INSERTED SUCCESSFULLY');
END;
```

```
END;
/
DECLARE
V_ROLLNO    NUMBER    := :ROLLNO;
V_ISSUEDATE DATE;
V_STATUS    VARCHAR2(10);
V_DAYS      NUMBER;
V_AMOUNT    NUMBER := 0;
BEGIN
SELECT ISSUEDATE, STATUS
INTO V_ISSUEDATE, V_STATUS
FROM BORROWERSS
WHERE ROLLNO = V_ROLLNO;

IF V_STATUS = 'ISSUED' THEN

V_DAYS := TRUNC(SYSDATE - V_ISSUEDATE);

IF V_DAYS BETWEEN 15 AND 30 THEN
V_AMOUNT := V_DAYS * 5;

ELSIF V_DAYS > 30 THEN
V_AMOUNT := V_DAYS * 50;

END IF;

UPDATE BORROWERSS
```

```
WHERE ROLLNO = V_ROLLNO,

IF V_STATUS = 'ISSUED' THEN

V_DAYS := TRUNC(SYSDATE - V_ISSUEDATE);

IF V_DAYS BETWEEN 15 AND 30 THEN
V_AMOUNT := V_DAYS * 5;

ELSIF V_DAYS > 30 THEN
V_AMOUNT := V_DAYS * 50;

END IF;

UPDATE BORROWERSS
SET STATUS = 'RETURN'
WHERE ROLLNO = V_ROLLNO;

INSERT INTO FINESS (ROLLNO, DATES, AMOUNT)
VALUES (V_ROLLNO, SYSDATE, V_AMOUNT);

DBMS_OUTPUT.PUT_LINE('FINE AMOUNT: ' || V_AMOUNT);
END IF;

END;
/
SELECT * FROM BORROWERSS;
SELECT * FROM FINESS;
```

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```
DATES DATE;  
AMOUNT NUMBER  
);  
DECLARE  
  V_ROLLNO    NUMBER      := :ROLLNO;  
  V_NAME      VARCHAR2(30) := :NAME;  
  V_ISSUEDATE DATE         := TO_DATE(:ISSUEDATE, 'DD-MM-YYYY');  
  V_BOOK      VARCHAR2(30) := :BOOK;  
  V_STATUS    VARCHAR2(10) := :STATUS;  
BEGIN  
  INSERT INTO BORROWERS (ROLLNO, NAME, ISSUEDATE, BOOK, STATUS)  
  VALUES (V_ROLLNO, V_NAME, V_ISSUEDATE, V_BOOK, V_STATUS);  
  DBMS_OUTPUT.PUT_LINE('RECORD INSERTED SUCCESSFULLY');  
END;  
/  
DECLARE  
  V_ROLLNO    NUMBER      := :ROLLNO;  
  V_ISSUEDATE DATE;  
  V_STATUS    VARCHAR2(10);
```

Results Explain Describe Saved SQL History

Enter Bind Variables - Brave

127.0.0.1:8080/apex/f?p=4500:138:3452043006940081::

Submit

:ROLLNO 68
:NAME Sahil Khan
:ISSUEDATE 18022026
:BOOK Data Comm
:STATUS ISSUED

Results Explain Describe Saved SQL History

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Autocommit Display 10 Save Run

```
IF V_DAYS BETWEEN 15 AND 30 THEN  
  V_AMOUNT := V_DAYS * 5;  
ELSEIF V_DAYS > 30 THEN  
  V_AMOUNT := V_DAYS * 50;  
END IF;  
UPDATE BORROWERS  
SET STATUS = 'RETURN'  
WHERE ROLLNO = V_ROLLNO;  
INSERT INTO FINESS (ROLLNO, DATES, AMOUNT)  
VALUES (V_ROLLNO, SYSDATE, V_AMOUNT);  
DBMS_OUTPUT.PUT_LINE('FINE AMOUNT: ' || V_AMOUNT);  
END IF;  
END;  
SELECT * FROM BORROWERS;  
SELECT * FROM FINESS;
```

Results Explain Describe Saved SQL History

RECORD INSERTED SUCCESSFULLY

Statement processed.

0.00 seconds

Language: en-us

Application Express 2.1.0.00.39
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Enter Bind Variables - Brave

127.0.0.1:8080/apex/f?p=4500:138:3452043006940081::

Submit

:ROLLNO 63

```
END;
/
SELECT * FROM BORROWERSS;
SELECT * FROM FINISS;
```

Results Explain Describe Saved SQL History

ROLLNO	NAME	ISSUEDATE	BOOK	STATUS
72	Sidra Khan	13-FEB-26	Software engg	RETURN
68	Sahil Khan	18-FEB-26	Data Comm	ISSUED
3	Gayatri Dhakne	23-MAR-26	PHP	RETURN
73	Satyam Singh	02-MAR-26	Operating System	ISSUED
37	Om Dhole	08-JAN-26	Machine Learning	ISSUED
56	Saipriya Khedlekar	29-JAN-26	System Software	ISSUED
1	Kashish Sonawane	02-FEB-26	Java Programming	ISSUED
63	Rida Patel	02-MAR-26	Python Programming	ISSUED

8 rows returned in 0.00 seconds [CSV Export](#)

```
SELECT * FROM BORROWERSS;
SELECT * FROM FINISS;
```

Results Explain Describe Saved SQL History

ROLLNO	DATES	AMOUNT
63	20-MAR-26	300
72	04-MAR-26	95
63	04-MAR-26	0

3 rows returned in 0.00 seconds [CSV Export](#)

For deleting a record from pl/sql table :-

the query is : DELETE FROM TABLE_NAME WHERE COLUMN_NAME = 'VALUE';

eg.

DELETE FROM BORROWERSS WHERE ROLLNO = 63;